

Unit 6

IIR Filter Design

1. IIR Filter Design: IIR filter design by classical filter design using low pass approximations Butterworth, Chebychev, Inverse Chebyshev, Elliptic and Bessel-Thompson filters
2. IIR filter design by Impulse-invariant method, Bilinear Transformation Method, Matched z-transform method
3. IIR lowpass discrete filter design using bilinear transformation
4. Spectral transformations, Highpass, Bandpass and Notch filters

Note: Teaching Hours: **(5 hours)**

Basic Analog Filter Approximations

It is required as practical characteristics of a filter is not identical to the characteristics of an ideal filter. There are

Three different types of approximation techniques. They are:

- Butterworth filter approximation
- Chebyshev filter approximation
- Elliptic filter approximation

Butterworth Filter approximation

The typical characteristics of a Butterworth low-pass filter is shown in figure below:

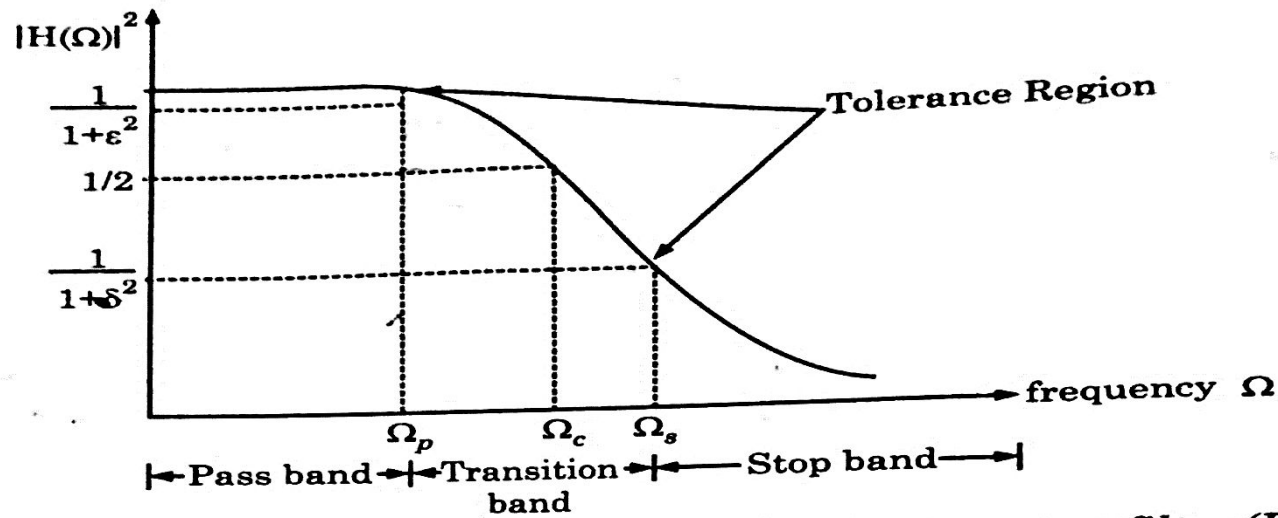


FIGURE 8.14 Typical characteristics of analog low-pass filter (LPF).

This type of response is called as Butterworth response because its main characteristic is that the passband is maximally flat. This means that there are no variations (ripples) in the passband. Now, the magnitude squared response of low pass Butterworth filter is given by

$$|H(\Omega)|^2 = \frac{1}{1 + \left(\frac{\Omega}{\Omega_c}\right)^{2N}} \quad \dots(8.53)$$

This equation may also be expressed as,

$$|H(\Omega)|^2 = \frac{1}{1 + \epsilon^2 \left(\frac{\Omega}{\Omega_p}\right)^{2N}} \quad \dots(8.54)$$

Here, $|H(\Omega)|$ = Magnitude of analog low pass filter (LPF).

Ω_c = Cut-off frequency (– 3 dB frequency).

Ω_p = Passband edge frequency.

$1 + \epsilon^2$ = Passband edge value.

$1 + \delta^2$ = Stopband edge value.

ϵ = Parameter related to ripples in passband.

δ = Parameter related to ripples in stopband.

N = Order of the filter.

We know that, in case of low pass filter (LPF), the frequencies will pass upto the value of cut-off frequency (Ω_c). This is called as passband. After that the frequencies are attenuated. This is called as stop band. Ideal characteristic is shown by dotted line in figure 8.15. Ideally, at the value of cut-off frequency (Ω_c) the frequencies should be stopped. However, in practical cases this is not happening.

Now the order of filter is denoted by N . Roughly we can say order of filter means, the number of stages used in the design of analog filter. As the order of filter N increases, the response of filter is more close to the ideal response as shown in figure 8.15.

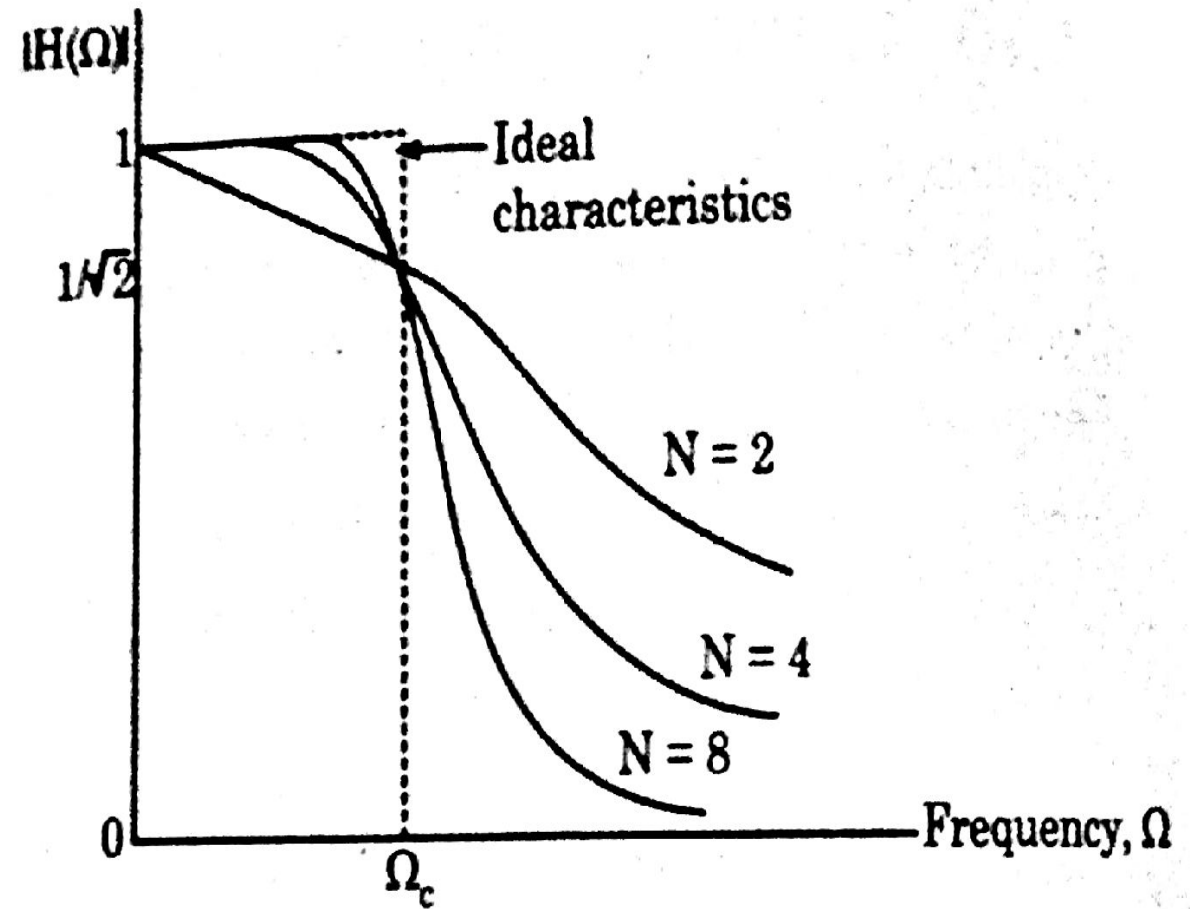


FIGURE 8.15 *Effect of N on frequency response characteristics.*

8.11.2 Designing Expressions and Designing Steps

Let A_p = Attenuation in passband.
 A_s = Attenuation in stopband.
 Ω_p = Pass band edge frequency.
 Ω_c = Cut-off frequency
 Ω_s = Stopband edge frequency.

In numerical problems, the specifications of required digital filter is usually given and it is asked to design a particular discrete time Butterworth filter. Then the following steps must be used :

1. From the given specifications of digital filter, we obtain equivalent analog filter as under :

(a) For *impulse invariance method*, we have

$$\Omega = \frac{\omega}{T} \quad |$$

(b) For *bilinear transformation method*, we have

$$\Omega = \frac{2}{T} \tan \frac{\omega}{2}$$

Here, Ω = Frequency of analog filter
 ω = Frequency of digital filter
 T = Sampling time

2. We evaluate the order N of filter using the following expression:

$$N = \frac{1}{2} \times \frac{\log \left[\frac{\left(\frac{1}{A_s^2} - 1 \right)}{\left(\frac{1}{A_p^2} - 1 \right)} \right]}{\log \left(\frac{\Omega_s}{\Omega_p} \right)} = \frac{\log \left[\left(1/\delta_s^2 \right) - 1 \right]}{2 \log (\Omega_s / \Omega_c)}$$

Here, δ_s = Attenuation in stop band.

If the specifications are given in decibels (dB) then, we make use of the following expression:

$$N = \frac{1}{2} \frac{\log \left[\frac{10^{0.1 A_s(\text{dB})} - 1}{10^{0.1 A_p(\text{dB})} - 1} \right]}{\log \left(\frac{\Omega_s}{\Omega_p} \right)}$$

3. After that we determine cut-off frequency (Ω_c).

The cut-off frequency (Ω_c) of analog filter is calculated as under :

(a) For *impulse invariance method*, we have

$$\Omega_c = \frac{\omega_c}{T}$$

(b) For bilinear transformation method, we have

$$\Omega_c = \frac{2}{T} \tan \frac{\omega_c}{2}$$

In case, ω_c is not given then we the following expression:

(i) $\Omega_c = \frac{\Omega_p}{\left(\frac{1}{A_p^2} - 1 \right)^{1/2N}}$ and if specifications are in dB then

(ii) $\Omega_c = \frac{\Omega_p}{\left[10^{0.1 A_p} - 1 \right]^{1/2N}}$

4. Next, we determine the poles using the following expression :

$$p_i = \pm \Omega_c e^{j(N+1)\pi/2N}, i = 0, 1, 2, \dots, N-1.$$

If the poles are complex conjugate then organize the poles (p_i) as complex conjugate pairs that means, s_1 and s_1^* , s_2 and s_2^* etc.

5. Next, we calculate the system transfer function of analog filter using following expression:

$$H_a(s) = \frac{\Omega_c^N}{(s - p_1)(s - p_2) \dots}$$

and if poles are complex conjugate then, we have

$$H_a(s) = \frac{\Omega_c^N}{(s - s_1)(s - s_1^*)(s - s_2)(s - s_2^*)}$$

6. Lastly, we design the digital filter using impulse invariance method or bilinear transformation method.

EXAMPLE 8.20 A digital filter has following frequency specification:

Passband frequency = $\omega_p = 0.2 \pi$

Stopband frequency = $\omega_s = 0.3 \pi$

What are the corresponding specifications for passband and stopband frequencies in analog domain if,

(i) Impulse invariance technique is used for designing.

(ii) Bilinear transformation is used for designing.

Solution: Here, let us assume sampling time $T = 1$ sec

(i) For impulse invariance method, we have

$$\Omega_p = \frac{\omega_p}{T} = 0.2 \pi = 0.63 \text{ rad/sec.}$$

and

$$\Omega_s = \frac{\omega_s}{T} = \frac{0.3\pi}{T} = 0.3\pi = 0.94 \text{ rad/sec.}$$

(ii) For bilinear transformation, we have

$$\begin{aligned}\Omega_p &= \frac{2}{T} \tan \left(\frac{\omega_p}{2} \right) \\ &= 2 \tan \left(\frac{0.2\pi}{2} \right) = 2 \tan \left(\frac{0.2 \times 180}{2} \right) = 0.65 \text{ rad/sec.}\end{aligned}$$

and
$$\Omega_s = \frac{2}{T} \tan\left(\frac{\omega_s}{2}\right)$$

or
$$\Omega_s = 2 \tan\left(\frac{0.3\pi}{2}\right) = 2 \tan\left(\frac{0.3 \times 180}{2}\right) = 1.019 \text{ rad/sec.} \quad \text{Ans.}$$

EXAMPLE 8.21 Design a second order discrete-time Butterworth filter with cut-off frequency of 1 kHz and sampling frequency of 10^4 samples/sec by bilinear transformation.

Solution: In this problem, the specifications of digital filter are not given directly. So, first we have to obtain required design specifications for digital filter. Then, for Butterworth approximation, we have to convert the specifications into specifications of equivalent analog filter. Finally, using bilinear transformation we have to obtain $H(z)$.

Given that

Order of filter, $N = 2$

Cut-off frequency of analog filter, $F_c = 1 \text{ kHz} = 1000 \text{ Hz}$.

Sampling frequency, $F_s = 10^4 \text{ samples/sec} = 10,000 \text{ Hz}$.

First, let us determine the required design specifications of *digital filter*.

We have the equation to convert continuous frequency (F) into discrete frequency (f). It is,

$$f = \frac{F}{F_s}$$

Thus,
$$f_c = \frac{F_c}{F_s} = \frac{1000}{10,000} = 0.1 \text{ cycles/sample}$$

Now, the angular frequency (frequency of digital filter) is

$$\omega_c = 2\pi f_c = 2\pi \times 0.1 = 0.2\pi \text{ radians/sample}$$

Now, let us determine the specifications of analog filter for *Butterworth approximation*.

For bilinear transformation, we have

$$\Omega = \frac{2}{T} \tan \frac{\omega}{2}$$

Thus,
$$\Omega_c = \frac{2}{T} \tan \frac{\omega_c}{2}$$

Here $T = \text{sampling time} = \frac{1}{F_s} = \frac{1}{10,000}$

or
$$\Omega_c = (2 \times 10,000) \tan\left(\frac{0.2\pi}{2}\right) = 6498.39 \text{ radians/sec.}$$

The value of N is given so it is not necessary to calculate it. Thus, the specifications of analog filter are as under:

(i) Cut-off frequency = $\Omega_c = 6498.39$ radian/sec.

(ii) Order of filter = $N = 2$.

Let us Calculate the poles using,

$$p_i = \Omega_c e^{j(N+2i+1)\pi/2N} \dots i = 0, 1, \dots N-1$$

Here $N = 2$, Thus, $i = 0$ to $N-1$ means $i = 0$ and $i = 1$.

(i) For $i = 0$, we have

$$\begin{aligned} p_0 &= \pm \Omega_c e^{j(N+1)\pi/2N} = \pm 6498.39 e^{j3\pi/4} = \pm 6498.39 \left[\cos\left(\frac{3\pi}{4}\right) + j \sin\left(\frac{3\pi}{4}\right) \right] \\ &= \pm 6498.39 [-0.707 + j 0.707] \end{aligned}$$

or $p_0 = 6498.39 [-0.707 - j 0.707]$ and $-6498.39 [-0.707 + j 0.707]$

or $p_0 = -4595.05 + j 4595.05$ and $4595.05 - j 4595.05$

(ii) For $i = 1$, we have

$$p_1 = \pm \Omega_c e^{j(N+2+1)\pi/2N}$$

$$= \pm 6498.39 e^{j(5\pi/4)}$$

or $p_1 = \pm 6498.39 \left[\cos\left(\frac{5\pi}{4}\right) + j \sin\left(\frac{5\pi}{4}\right) \right]$

$$= \pm 6498.39 [-0.707 - j 0.707]$$

or $p_1 = 6498.39 [-0.707 - j 0.707]$

and $-6498.39 [-0.707 - j 0.707]$

or $p_1 = -4595.05 - j 4595.05$

and $4595.05 + j 4595.05$

Now, let us plot these poles as shown in figure 8.16.

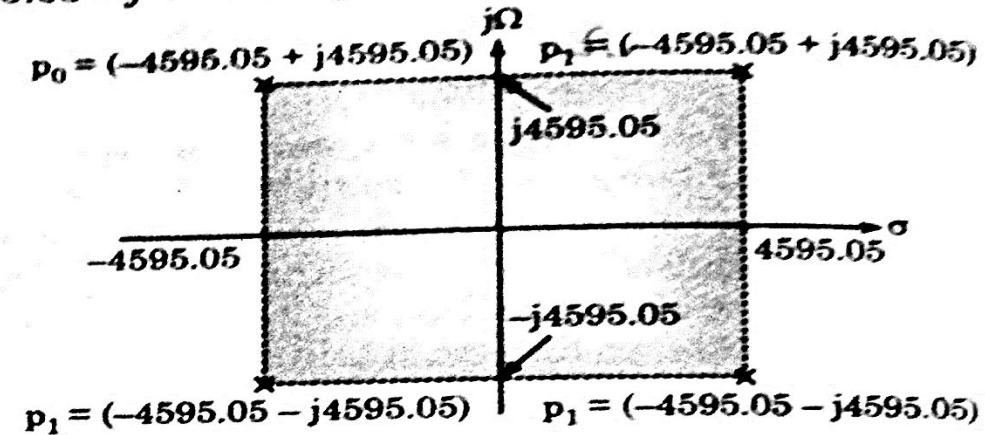


FIGURE 8.16 Plot of poles, p_i

Note : We know that analog filter is stable if the poles lie on the L.H.S. of s -plane. So for the filter to be stable, we have to select the poles which are on the L.H.S. of s -plane. Observe figure 8.16. We shall denote these poles by s_1 and s_1^* ; because these poles are complex conjugate of each other.

Thus, $s_1 = -4595.05 + j 4595.05$

and $s_1^* = -4595.05 - j 4595.05$

Now, the transfer function of analog filter is obtained by using the following expression:

$$H_a(s) = \frac{\Omega_c^N}{(s - s_1)(s - s_1^*)} = \frac{(6498.39)^2}{(s + 4595.05 - j 4595.05)(s + 4595.05 + j 4595.05)}$$

Simplifying, we get

$$H_a(s) = \frac{(6498.39)^2}{s^2 + 9190.1s + 42.22 \times 10^6}$$

Lastly, let us design digital filter using bilinear transformation.

In case of bilinear transformation, $H(z)$ is obtained by putting $s = \frac{2}{T} \left[\frac{z-1}{z+1} \right]$ in equation of $H_a(s)$

Here, $T = \frac{1}{10,000}$.

Thus, we have to put,

$$s = 2 \times 10^4 \left[\frac{z-1}{z+1} \right] \text{ in the equation of } H_a(s)$$

Therefore,

$$H(z) = \frac{(6498.39)^2}{\left[2 \times 10^4 \left(\frac{z-1}{z+1} \right) \right]^2 + 183.802 \times 10^6 \left(\frac{z-1}{z+1} \right) + 42.22 \times 10^6}$$

This is the required transfer function for digital filter. **Ans.**

EXAMPLE 8.22 Using bilinear transformation, design a Butterworth filter which satisfies the following conditions :

$$0.8 \leq |H(e^{j\omega})| \leq 1$$

$$0 \leq \omega \leq 0.2\pi$$

$$|H(e^{j\omega})| \leq 0.2,$$

$$0.6\pi \leq \omega \leq \pi$$

Solution: $|H(e^{j\omega})|$ is same as $|H(\omega)|$. Therefore, the specifications are given for the required digital filter. These specifications are shown in figure 8.17.

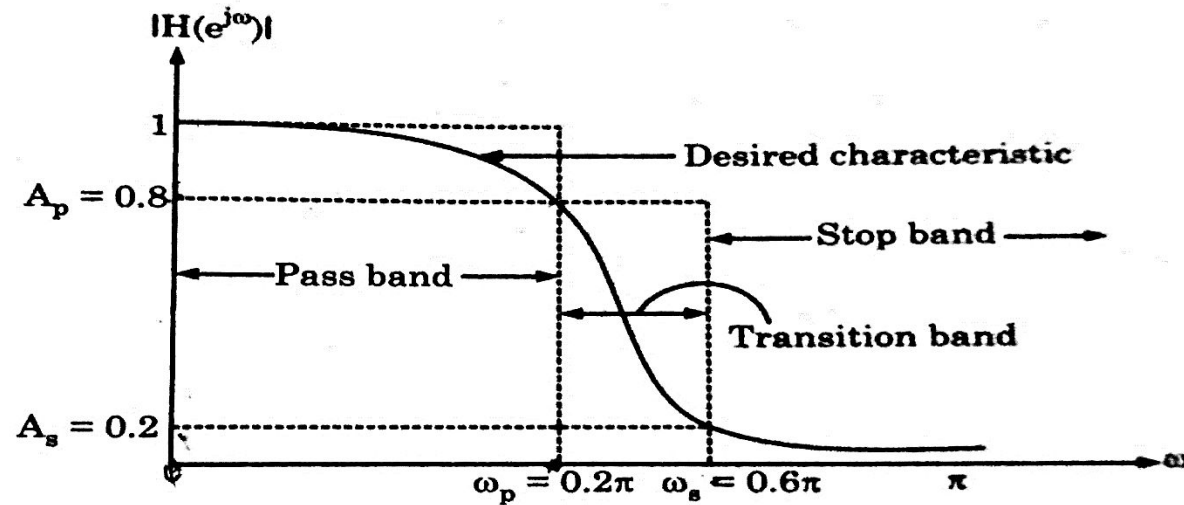


FIGURE 8.17 Characteristic plot for digital filter.

Given that

$$A_p = 0.8,$$

$$\omega_p = 0.2\pi$$

$$A_s = 0.2,$$

$$\omega_s = 0.6\pi$$

Part A : Determination of specifications of analog filter

(i) We shall obtain the values of Ω_p and Ω_s . We have to use bilinear transformation method. Hence, for bilinear transformation, we have

$$\Omega = \frac{2}{T} \tan \frac{\omega}{2}$$

or

$$\Omega_p = \frac{2}{T} \tan \frac{\omega_p}{2}$$

The value of sampling time T is not given. So assume $T = 1$.

Thus,
$$\Omega_p = 2 \tan \frac{0.2\pi}{2} = 0.65$$

Similarly,
$$\Omega_s = \frac{2}{T} \tan \frac{\omega_p}{2} = 2 \tan \frac{0.6\pi}{2} = 2.75$$

Therefore, specifications of analog filter are as under :

$$A_p = 0.8, \quad \Omega_p = 0.65$$

$$A_s = 0.2, \quad \Omega_s = 2.75$$

(ii) Let us determine the order of filter, N .

It is expressed as

$$N = \frac{1}{2} \times \frac{\log \left[\left(\frac{1}{A_s^2} - 1 \right) / \left(\frac{1}{A_p^2} - 1 \right) \right]}{\log \frac{\Omega_s}{\Omega_p}} = \frac{1}{2} \times \frac{\log \left[\left(\frac{1}{(0.2)^2} - 1 \right) / \left(\frac{1}{(0.8)^2} - 1 \right) \right]}{\log \left(\frac{2.75}{0.65} \right)}$$

or
$$N = \frac{1}{2} \times \frac{\log \left[\frac{24}{0.5625} \right]}{\log \left(\frac{2.75}{0.65} \right)} = 1.3$$

Thus, approximately, we can take $N = 2$.

(iii) Next, let us determine the cut-off frequency, Ω_c

Here, ω_c is not given, so, let us use the following expression:

$$\Omega_c = \frac{\Omega_p}{\left(\frac{1}{A_p^2} - 1 \right)^{1/2N}} = \frac{0.65}{\left(\frac{1}{(0.8)^2} - 1 \right)^{1/4}} = 0.75$$

(iv) Now, let us determine the poles of $H_a(s)$.

We know that the poles of $H_a(s)$ are given by

$$p_i = \pm \Omega_c e^{j(N+2i+1)\pi/2N}, i = 0, 1, 2, \dots, N-1$$

or

$$p_i = \pm 0.75 e^{j(2+2i+1)\pi/4} = \pm 0.75 e^{j(3+2i)\pi/4}$$

Here, range of i is from 0 to $N-1$ and $N=2$. Hence, the value of i is, $i=0$ and $i=1$.

For $i=0$, we have

$$p_0 = \pm 0.75 e^{j(3)\pi/4} = \pm 0.75 \left[\cos\left(\frac{3\pi}{4}\right) + j \sin\left(\frac{3\pi}{4}\right) \right]$$

or $p_0 = \pm 0.75 [-0.707 + j 0.707]$

or $p_0 = \pm [-0.53 + j 0.53]$

or $p_0 = -0.53 + j 0.53$ and $0.53 - j 0.53$

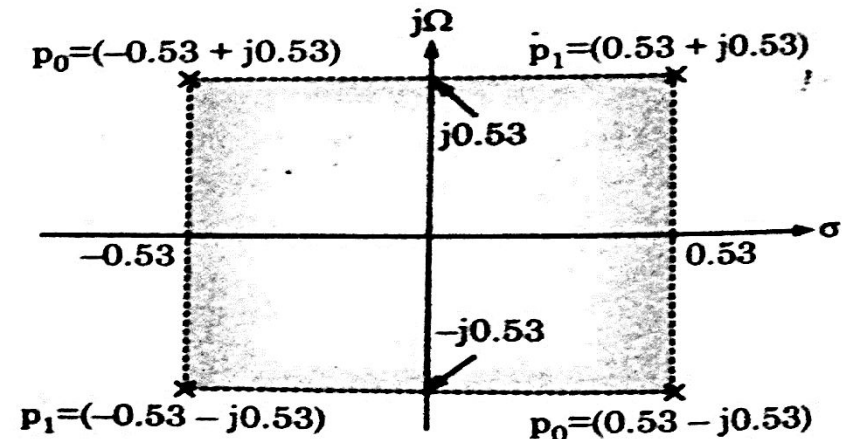
For $i=1$, we have

$$p_1 = \pm 0.75 e^{j(3+2)\pi/4} = \pm 0.75 e^{j5\pi/4}$$

or $p_1 = \pm 0.75 \left[\cos\left(\frac{5\pi}{4}\right) + j \sin\left(\frac{5\pi}{4}\right) \right]$

or $p_1 = \pm 0.75 [-0.707 - j 0.707]$

or $p_1 = -0.53 - j 0.53$ and $0.53 + j 0.53$



(v) Now, let us determine $H_a(s)$.

The transfer function of analog filter is given by

$$H_a(s) = \frac{\Omega_c^N}{(s - s_1)(s - s_1^*)} = \frac{(0.75)^2}{(s + 0.53 - j0.53)(s + 0.53 + j0.53)}$$

or

$$H_a(s) = \frac{(0.75)^2}{s^2 + 0.53s + j0.53s + 0.53s + 0.28 + j0.28 - j0.53s - j0.28 + 0.28}$$

or

$$H_a(s) = \frac{(0.75)^2}{s^2 + 1.06s + 0.56} = \frac{0.56}{s^2 + 1.06s + 0.56}$$

Part B : Determination of transfer function for digital filter

We have to use bilinear transformation method. Then, $H(z)$ is obtained by putting $s = \frac{2}{T} \left(\frac{z-1}{z+1} \right)$ in the equation of $H_a(s)$.

Let us assume sampling time $T = 1$.

Therefore, we have

$$H(z) = \frac{0.56}{4\left(\frac{z-1}{z+1}\right)^2 + (2 \times 1.06)\left(\frac{z-1}{z+1}\right) + 0.56} = \frac{0.56}{\frac{4(z-1)^2}{(z+1)^2} + \frac{2.12z - 2.12}{z+1} + 0.56}$$

or

$$H(z) = \frac{0.56(z+1)^2}{4(z^2 - 2z + 1) + (2.12z - 2.12)(z+1) + 0.56(z+1)^2}$$

or

$$H(z) = \frac{0.56(z+1)^2}{4z^2 - 8z + 4 + 2.12z^2 + 2.12z - 2.12z - 2.12 + 0.56z^2 + 1.12z + 0.56}$$

or

$$H(z) = \frac{0.56(z+1)^2}{6.68z^2 - 6.88z + 2.44} \quad \eta$$

This is the required transfer function to design the digital filter. **Ans.**

8.12 CHEBYSHEV FILTER APPROXIMATION

Using Chebyshev filter design, there are two subgroups as under:

- (i) Type-1 Chebyshev filter
- (ii) Type-2 Chebyshev filter

8.12.1 Type-1 Chebyshev Filter

These filters are all pole filters. In the passband, these filters show equiripple behaviour and they have monotonic characteristics in the stopband.

The filter characteristics for odd and even values of N are shown in figure 8.20.

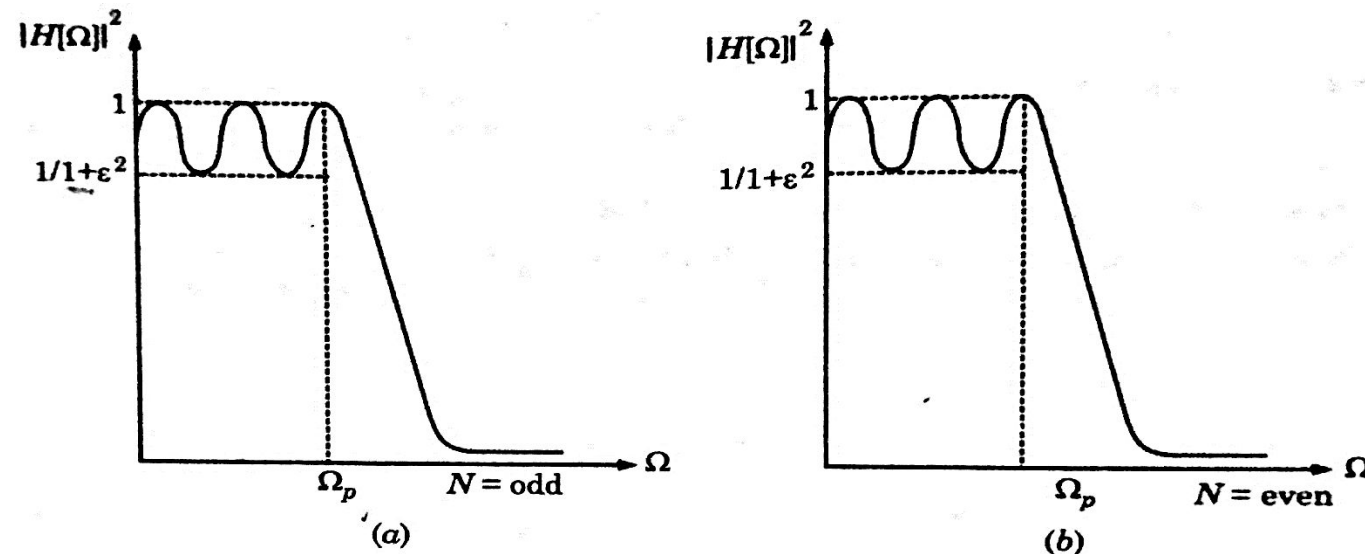


FIGURE 8.20 Type-1 Chebyshev filter characteristics.

The magnitude squared frequency response is given by

$$|H(\Omega)|^2 = \frac{1}{1 + \epsilon^2 C_N^2(\Omega / \Omega_p)} \quad \dots(8.55)$$

Here, ϵ = Ripple parameter in the passband

Ω_p = Passband frequency

$C_N(x)$ = Chebyshev polynomial of order N .

The chebyshev polynomials are determined by using the following

$$C_{N+1}(x) = 2xC_N(x) - C_{N-1}(x)$$

...(8.56)

with

$$C_0(x) = 1 \text{ and } C_1(x) = x$$

The different polynomials are given in Table 8.1.

Table 8.1. Different Polynomials

S. No.	N	$C_N(x)$
1	0	1
2	1	x
3	2	$2x^2 - 1$
4	3	$4x^3 - 3x$
5	4	$8x^4 - 8x^2 + 1$
6	5	$16x^5 - 20x^3 + 5x$

The major difference between Butterworth and Chebyshev filter is that the poles of Butterworth filter lie on the circle, whereas the poles of Chebyshev filter lie on ellipse.

8.12.2 Type-II Chebyshev Filter

This filter consists of zeros as well as poles. The characteristic of such filters for even and odd values of N is shown in figure 8.21.

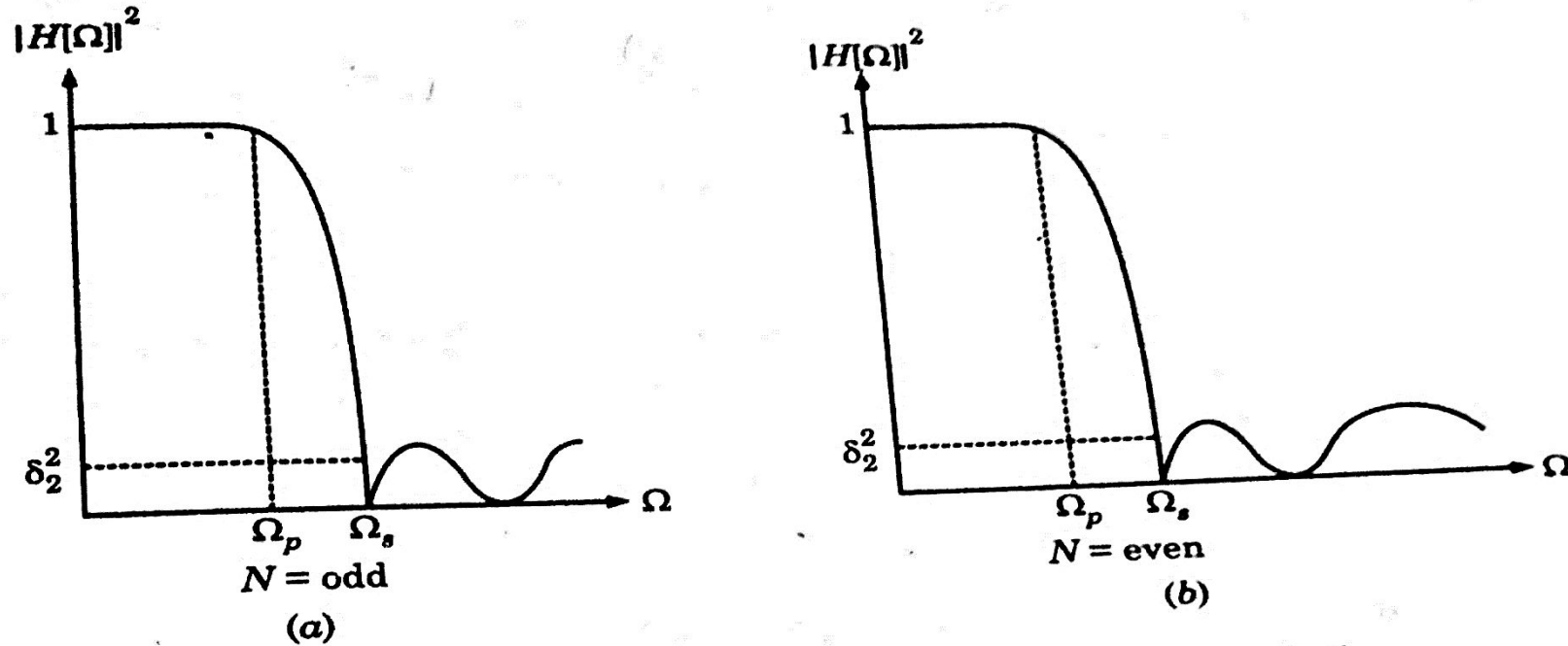


FIGURE 8.21 Type-II Chebyshev filter characteristic.

The magnitude squared response is given by

$$|H(\Omega)|^2 = \frac{1}{1 + \epsilon^2 \left[C_N^2 \left(\frac{\Omega_s}{\Omega_p} \right) / C_N^2 \left(\frac{\Omega_s}{\Omega} \right) \right]}$$

Here $C_N(x)$ = N^{th} order polynomial
 Ω_s = Stopband frequency
 Ω_p = Passband frequency

8.12.3 Chebyshev Low Pass Filter Design

(i) Frequency response

The magnitude squared frequency response of chebyshev filter is given by

$$|H(j\Omega)|^2 = \frac{1}{1 + \epsilon^2 C_N^2\left(\frac{\Omega}{\Omega_p}\right)}$$

Here, $C_N\left(\frac{\Omega}{\Omega_p}\right)$ = Chebyshev polynomial of order M .

(ii) Parameter ϵ

It represents ripple parameter in the passband. It is given by

$$\epsilon = [10^{0.1 A_p \text{ dB}} - 1]^{1/2}$$

If A_p is not in dB then ϵ is calculated using the following expression:

$$\epsilon = \left[\frac{1}{A_p^2} - 1 \right]^{1/2}$$

(iii) At cut-off frequency $\Omega_c = 1$, the magnitude is given by,

$$|H(j\Omega)| = \frac{1}{\sqrt{1 + \epsilon^2}}$$

(iv) Order of a filter

When magnitude is expressed in dB, then order (N) of a filter is obtained by using the following expression:

$$|H(j\Omega)| \text{ in dB} = -20 \log_{10} \epsilon - 6(N-1) - 20 \log(\Omega_s)$$

(v) Position of poles of Chebyshev filter

First, let us calculate regular position of poles of Butterworth filter using

$$\theta_i = \frac{\pi}{2} + \frac{(2i+1)\pi}{2N}, \quad i = 0, 1, 2, \dots, N-1$$

The position of poles of Chebyshev filter lie on ellipse at co-ordinates x_i and y_i given by,

$$x_i = r \cos \theta_i, \quad i = 0, 1, 2, \dots, N-1$$

$$y_i = R \sin \theta_i, \quad i = 0, 1, 2, \dots, N-1$$

Here r represents minor axis of ellipse and is given by,

$$r = \Omega_p \frac{\beta^2 - 1}{2\beta}$$

and R represents major axis of ellipse and is given by,

$$R = \Omega_p \frac{\beta^2 + 1}{2\beta}$$

(v) Position of poles of Chebyshev filter

First, let us calculate regular position of poles of Butterworth filter using

$$\theta_i = \frac{\pi}{2} + \frac{(2i+1)\pi}{2N}, \quad i = 0, 1, 2, \dots, N-1$$

The position of poles of Chebyshev filter lie on ellipse at co-ordinates x_i and y_i given by

$$x_i = r \cos \theta_i, \quad i = 0, 1, 2, \dots, N-1$$

$$y_i = R \sin \theta_i, \quad i = 0, 1, 2, \dots, N-1$$

Here r represents minor axis of ellipse and is given by,

$$r = \Omega_p \frac{\beta^2 - 1}{2\beta}$$

and R represents major axis of ellipse and is given by,

$$R = \Omega_p \frac{\beta^2 + 1}{2\beta}$$

Here, the parameter β is given by,

$$\beta = \left[\frac{\sqrt{1 + \epsilon^2} + 1}{\epsilon} \right]^{\frac{1}{N}}$$

Then, the pole positions are denoted by s_p , and

$$s_p = r \cos \theta_i + jR \sin \theta_i$$

(vi) System transfer function

The system transfer function of analog filter is given by,

$$H_a(s) = \frac{i}{(s - s_0)(s - s_1)(s - s_2) \dots}$$

After simplification this equation can be written as,

$$H_a(s) = \frac{i}{s^N + b_{N-1}s^{N-1} + \dots + b_0}$$

Here, b_0 = Constant term in the denominator

Now, the value of i is calculated as under:

$$i = \begin{cases} b_0 & \text{for } N \text{ odd} \\ \frac{b_0}{\sqrt{1 + \epsilon^2}} & \text{for } N \text{ even} \end{cases}$$

EXAMPLE. 8.25 Design a low-pass 1 rad/sec bandwidth Chebyshev filter with following characteristics.

- (i) Acceptable passband ripple of 2 dB.
- (ii) Cut-off radian frequency of 1 rad/sec.
- (iii) Stopband attenuation of 20 dB or greater beyond 1.3 rad/sec.

Solution: Given data

Given cut-off frequency is 1 rad/sec. This means that it is normalized low pass chebyshev filter.

$$\text{Passband ripple} = A_p = 2 \text{ dB}$$

$$\text{Stopband attenuation} = A_s = 20 \text{ dB}$$

$$\text{Stopband frequency} = \Omega_s = 1.3 \text{ rad/sec.}$$

First, let us determine the parameter ϵ .

We have,

$$\epsilon = \left[10^{0.1A_p \text{ dB}} - 1 \right]^{\frac{1}{2}} = \left[10^{0.1(2)} - 1 \right]^{\frac{1}{2}} = 0.765$$

$$A_p = 2 \text{ dB} \Rightarrow \frac{1}{1 + \epsilon^2} = \frac{1}{1 + 0.765^2} = \frac{1}{1.586} \approx 0.631$$

Next, we determine the order N of filter.

Given stopband attenuation = 20 dB for $\Omega_s = 1.3$ rad/sec.

Since, it is attenuation, therefore, we can write

$$|H(j\Omega)| = -20 \text{ dB for } \Omega = \Omega_s = 1.3 \text{ rad/sec.}$$

The order N of filter is calculated by using the following expression:

$$|H(j\Omega)| \text{ in dB} = -20 \log_{10} \epsilon - 6(N-1) - 20N \log_{10} \Omega_s$$

$$\text{or } -20 = -20 \log_{10} (0.765) - 6(N-1) - 20N \log_{10} (1.3)$$

$$\text{or } 6(N-1) + 20N \log_{10} 1.3 = 20 - 20 \log_{10} (0.765)$$

$$6N - 6 + 2.28N = 22.327$$

$$\text{or } N = 3.421$$

Since, N is order of filter, we can write

$$N \cong 4$$

Now, we shall determine the poles.

The pole positions are given by,

$$s_p = r \cos \theta_i + iR \sin \theta_i$$

...(i)

First, we will calculate parameter β , i.e.

$$\beta = \left[\frac{\sqrt{1+\epsilon^2} + 1}{\epsilon} \right]^{\frac{1}{N}} = \left[\frac{\sqrt{1+(0.765)^2} + 1}{0.765} \right]^{\frac{1}{4}} = 1.31 \quad \dots(ii)$$

Now, let us calculate values of r and R .

This is normalized filter. So, $\Omega = 1$ rad/sec.

We have minor axis of ellipse $= r = \Omega_p \cdot \frac{\beta^2 - 1}{2\beta}$

$$\text{or} \quad r = 1 \cdot \frac{(1.31)^2 - 1}{2(1.31)} = 0.273 \quad \dots(iii)$$

The major axis of an ellipse is given by,

$$R = \Omega_p \frac{\beta^2 + 1}{2\beta} = 1 \cdot \frac{(1.31)^2 + 1}{2(1.31)} = 1.04 \quad \dots(iv)$$

Now, we shall calculate values of θ_i as under :

$$\theta_i = \frac{\pi}{2} + \frac{(2i+1)\pi}{2N}, \quad i = 0, 1, 2, 3. \quad (0, 1, \dots, N-1)$$

$$\text{For } i = 0 : \quad \theta_0 = \frac{\pi}{2} + \frac{\pi}{8} = \frac{5\pi}{8}$$

$$\text{For } i = 1 : \quad \theta_1 = \frac{\pi}{2} + \frac{3\pi}{8} = \frac{7\pi}{8}$$

$$\text{For } i = 2 : \quad \theta_2 = \frac{\pi}{2} + \frac{5\pi}{8} = \frac{9\pi}{8}$$

$$\text{For } i = 3 : \quad \theta_3 = \frac{\pi}{2} + \frac{7\pi}{8} = \frac{11\pi}{8}$$

Using equation (i), we can calculate pole positions as under:

For $i = 0$: $s_0 = r \cos \theta_0 + jR \sin \theta_0$

or $s_0 = 0.273 \cos\left(\frac{5\pi}{8}\right) + j(1.04) \sin\left(\frac{5\pi}{8}\right) = -0.1 + j0.96$

For $i = 1$: $s_1 = r \cos \theta_1 + jR \sin \theta_1$

or $s_1 = 0.273 \cos\left(\frac{7\pi}{8}\right) + j(1.04) \sin\left(\frac{7\pi}{8}\right) = -0.25 + j0.4$

For $i = 2$: $s_2 = r \cos \theta_2 + jR \sin \theta_2$

or $s_2 = 0.273 \cos\left(\frac{9\pi}{8}\right) + j(1.04) \sin\left(\frac{9\pi}{8}\right) = -0.25 - j0.4$

For $i = 3$: $s_3 = r \cos \theta_3 + jR \sin \theta_3$

or $s_3 = 0.273 \cos\left(\frac{11\pi}{8}\right) + j(1.04) \sin\left(\frac{11\pi}{8}\right) = -0.1 - j0.96$

Lastly, we determine the system function $H_a(s)$.

The system function $H_a(s)$ is given by,

$$H_a(s) = \frac{i}{(s - s_0)(s - s_1)(s - s_2)(s - s_3)}$$

or
$$H_a(s) = \frac{i}{(s + 0.1 - j0.96)(s + 0.25 - j0.4)(s + 0.25 + j0.4)(s + 0.1 + j0.96)}$$

or
$$H_a(s) = \frac{i}{(s^2 + 0.2s + 0.9316)(s^2 + 0.5s + 0.2225)} = \frac{i}{s^4 + 0.7s^3 + 1.25s^2 + 0.51s + 0.207}$$

Here $N = 4$ which is even. Thus, we have

$$i = \frac{b_0}{\sqrt{1 + \epsilon^2}}$$

$$b_0 = \text{constant term in the denominator} = 0.207.$$

or
$$i = \frac{0.207}{\sqrt{1 + (0.765)^2}} = 0.16$$

Therefore, we have

$$H_a(s) = \frac{0.16}{s^4 + 0.7s^3 + 1.25s^2 + 0.51s + 0.207}$$

This is required transfer function.

Ans.

EXAMPLE. 8.27 Design a digital low pass filter using Chebyshev filter seeing procedure that meets the following specifications : Passband magnitude characteristics that is constant to within 1 dB for recurrences below $\omega = 0.2 \pi$ and stopband attenuation of at least 15 dB for frequencies between $\omega = 0.3 \pi$ and π . Use bilinear transformation.

Solution: In this problem, the specifications of digital filter are given. The given specifications are

$$\text{Passband ripple} = A_p = 1 \text{ dB}$$

$$\text{Passband edge frequency} = \omega_p = 0.2 \pi$$

$$\text{Stopband attenuation} = A_s = 15 \text{ dB}$$

$$\text{Stopband edge frequency} = \omega_s = 0.3 \pi$$

First, we will calculate the values of edge frequencies for the analog filter.

First, let us determine analog filter's edge frequencies.
For the bilinear transformation, we have

$$\Omega_p = \frac{2}{T_s} \tan \frac{\omega_p}{2}$$

⚡ **Note :** Sampling time is not given. Therefore, we will assume sampling time $T = 1$ sec.

Thus

$$\Omega_p = 2 \tan \frac{0.2\pi}{2} = 0.6498$$

and

$$\Omega_s = \frac{2}{T} \tan \frac{\omega_s}{2}$$

Again

$$T = 1 \text{ sec.}$$

Hence,

$$\Omega_s = 2 \tan \frac{0.3\pi}{2} = 1.019$$

Next, we determine the parameter ϵ .

It is given by,

$$\epsilon = \left[10^{0.1A_p \text{ dB}} - 1 \right]^{\frac{1}{2}} = \left[10^{0.1(1)} - 1 \right]^{\frac{1}{2}} = 0.5088$$

Now, let us determine the order (N) of a filter.

It is calculated using the equation,

$$|H(j\Omega)| \text{ in dB} = -20 \log_{10} \epsilon - 6(N-1) - 20N \log_{10}(\Omega_s)$$

Given attenuation is 15 dB.

$$\text{or} \quad -15 \text{ dB} = -20 \log_{10} 0.5088 - 6(N-1) - 20N \log_{10}(1.019)$$

$$\text{or} \quad -15 = 5.87 - 6N + 6 - 0.1634N$$

$$\text{or} \quad 6.1634N = 26.87$$

$$\text{or} \quad N = 4.35$$

Thus, order of filter = $N \approx 5$

The pole position are given by,

$$s_p = r \cos \theta_i + jR \sin \theta_i$$

...(i)

Now, we will calculate values of r , R and θ_i

$$\text{Minor axis of ellipse} = r = \Omega_p \frac{\beta^2 - 1}{2\beta}$$

and

$$\beta = \left[\frac{\sqrt{1+\epsilon^2} + 1}{\epsilon} \right]^{\frac{1}{N}} = \left[\frac{\sqrt{1+(0.5088)^2} + 1}{0.5088} \right]^{\frac{1}{5}} = 1.33$$

or

$$r = 0.6498 \cdot \frac{(1.33)^2 - 1}{2 \times 1.33} = 0.186$$

$$\text{Add major axis of ellipse} = R = \Omega_p \cdot \frac{\beta^2 + 1}{2\beta}$$

or

$$R = 0.6498 \frac{(1.33)^2 + 1}{2 \times 1.33} = 0.67$$

Now, we shall calculate values of θ_i
We have,

$$\theta_i = \frac{\pi}{2} + \frac{(2i+1)\pi}{2N}, \quad i = 0, 1, 2, \dots, N-1$$

that means $i = 0, 1, 2, 3, 4$

For $i = 0$: $\theta_i = \frac{\pi}{2} + \frac{\pi}{10} = \frac{6\pi}{10}$

For $i = 1$: $\theta_1 = \frac{\pi}{2} + \frac{(2+1)\pi}{10} = \frac{\pi}{2} + \frac{3\pi}{10} = \frac{8\pi}{10}$

For $i = 2$: $\theta_2 = \frac{\pi}{2} + \frac{5\pi}{10} = \pi$

For $i = 3$: $\theta_3 = \frac{\pi}{2} + \frac{7\pi}{10} = \frac{12\pi}{10}$

For $i = 4$: $\theta_4 = \frac{\pi}{2} + \frac{9\pi}{10} = \frac{14\pi}{10}$

Now putting these values in equation (i), we get

For $i = 0$

$$s_0 = 0.186 \cos \theta_0 + j0.67 \sin \theta_0$$

or

$$s_0 = 0.186 \cos \left(\frac{6\pi}{10} \right) + j0.67 \sin \left(\frac{6\pi}{10} \right) = -0.057 + j0.637$$

For $i = 2$

$$s_2 = 0.186 \cos \theta_2 + j0.67 \sin \theta_2$$

or

$$s_2 = 0.186 \cos \pi + j 0.67 \sin \pi = -0.186$$

For $i = 3$

$$s_3 = 0.186 \cos \theta_3 + j0.67 \sin \theta_3$$

or

$$s_3 = 0.186 \cos \left(\frac{12\pi}{10} \right) + j 0.67 \sin \left(\frac{12\pi}{10} \right) = -0.15 - j0.39$$

For $i = 4$

$$s_4 = 0.186 \cos \theta_4 + j0.67 \sin \theta_4$$

or

$$s_4 = 0.186 \cos \left(\frac{14\pi}{10} \right) + j 0.67 \sin \left(\frac{14\pi}{10} \right) = -0.057 - j0.639$$

Next, we determine the system transfer function of analog filter.

It is given by,

$$H_a(s) = \frac{i}{(s-s_0)(s-s_1)(s-s_2)(s-s_3)(s-s_4)}$$

or

$$H_a(s) = \frac{i}{(s+0.057-j0.637)(s-0.15-j0.39)(s+0.186)(s+0.15+j0.39)(s+0.057+j0.637)}$$

$$\text{or } H_a(s) = \frac{i}{(s+0.816)(s^2+0.3s+0.1746)(s^2+0.114s+0.409)}$$

Now, $N = 5$ thus,

$i = b_0 =$ Constant term in the denominator

$$\text{or } i = b_0 = (0.186)(0.1746)(0.409) = 0.0135$$

$$\text{or } H_a(s) = \frac{0.0132}{(s+0.186)(s^2+0.3s+0.1746)(s^2+0.114s+0.409)}$$

Lastly, we determine the transfer function for digital filter.

By using bilinear transformation, the transfer function for digital filter $H(z)$ is obtained from $H(s)$ by putting,

$$s = \frac{2}{T} \left[\frac{z-1}{z+1} \right]$$

Here, $T =$ Sampling time $= 1$ sec. (assumed)

$$\therefore H_a(z) = \frac{0.0132}{\left[2\left(\frac{z-1}{z+1}\right) + 0.186 \right] \left\{ \left[2\left(\frac{z-1}{z+1}\right) \right]^2 + 0.3 \left[2\left(\frac{z-1}{z+1}\right) \right] + 0.01746 \right\} \left\{ \left[2\left(\frac{z-1}{z+1}\right) \right]^2 + 0.114 \left[2\left(\frac{z-1}{z+1}\right) \right] + 0.409 \right\}}$$

8.13 ELLIPTIC FILTERS

Elliptic filters are also called as cauer filters. Such filters have equiripple passband and stopband. The magnitude squared frequency response of elliptic filter is shown in figure 8.22.

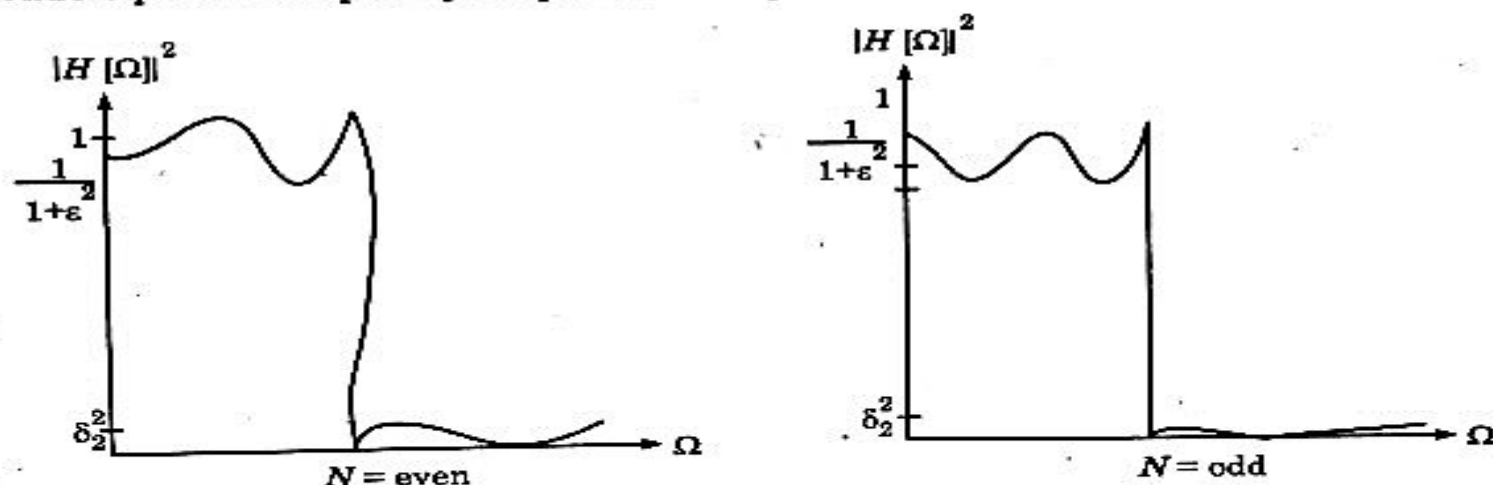


FIGURE 8.22 Elliptic filter characteristics.

The magnitude squared response of elliptic filter is obtained by using,

$$|H(\Omega)|^2 = \frac{1}{1 + \epsilon^2 U_N^2\left(\frac{\Omega}{\Omega_p}\right)} \quad \dots(8.57)$$

Here ϵ = Ripple parameter in passband
 Ω_p = Passband edge frequency
 $U_N(x)$ = Jacobian elliptic function of order N .

The order ' N ' of elliptic filter is given by,

$$N = \frac{K(\Omega_p/\Omega_s)K\left(\sqrt{1 - (\epsilon^2/\delta^2)}\right)}{K(\epsilon/\delta)K\left(\sqrt{1 - (\Omega_p/\Omega_s)^2}\right)}$$

Here δ is related to δ_1 and δ_2 as follows :

$$\delta_2 = \sqrt{1 + \delta^2} \quad \dots(8.58)$$

where δ_2 = stopband ripple

and $\delta_2 = 10 \log_{10} (1 + \epsilon^2)$

where δ_1 = stopband ripple

The function $K(x)$ is called as elliptic integral of first kind.

8.13.1 The Function $K(x)$ is Called as Elliptic Integral of First Kind

- (i) Elliptic filter is more efficient because it provides the smallest order filter for given set of specifications.
- (ii) It has smallest transition bandwidth.

8.13.2 Disadvantage of Elliptic Filters

The phase response is more non-linear in passband.

Unit 7

FIR Filter Design

8.9.1 Rectangular Window for FIR Filter Design

We know that the rectangular window is defined as:

$$w_R(n) = \begin{cases} 1 & \text{for } n = 0, 1, \dots, M-1 \\ 0 & \text{elsewhere} \end{cases} \quad \dots(8.56)$$

The unit sample response of the FIR filter is obtained by multiplying $h_d(n)$ by $w_R(n)$.

This means that $h(n) = h_d(n) w_R(n)$...(8.57)

From equation (8.56), substituting for $w_R(n)$ above expression will become,

$$h(n) = \begin{cases} h_d(n) & \text{for } n = 0, 1, \dots, M-1 \\ 0 & \text{elsewhere} \end{cases} \quad \dots(8.58)$$

Hence, $h_d(n)$ is truncated to length 'M' to give $h(n)$. This truncation is obtained by multiplication of $h_d(n)$ and $w_R(n)$. This is also called as **windowing function** $w_R(n)$ acts like a window for $h_d(n)$.

The Fourier transform of rectangular window is given by equation (8.41) and its magnitude response is given by equation (8.42), i.e.,

$$|W_R(\omega)| = \frac{\left| \sin\left(\frac{\omega M}{2}\right) \right|}{\left| \sin\left(\frac{\omega}{2}\right) \right|} \quad \dots(8.59)$$

Figure 8.7 shows the magnitude frequency response given by above expression.

Few Important Points

(i) From equation (8.57), we know that unit sample response of FIR filter is given as:

$$h(n) = h_d(n) w(n)$$

Here, $w(n)$ represents generalized window function.

(ii) The frequency response of FIR filter can be obtained by taking Fourier transform of above equation. This means that

$$H(\omega) = \text{FT} \{h_d(n) \cdot w(n)\}$$

(iii) We know that Fourier transform of multiplication of two signals is equal to the convolution of their individual Fourier transforms. Then, above equation will become:

$$H(\omega) = H_d(\omega) \otimes W(\omega) \quad \dots(8.60)$$

This expression shows that the response of FIR filter is equal to the convolution of desired frequency response with that of window function. Because of convolution, $H(\omega)$ has the smoothing effect. The sidelobes of $W(\omega)$ create undesirable ringing effects in $H(\omega)$.

8.9.2 Gibbs Phenomenon

(JNTU, Hyderabad, Sem. Exam; 2007-08, 2008-09)

Let us consider the example of a low-pass filter having desired frequency response $H_d(\omega)$ as depicted in figure 8.13 (a). This response has the cut-off frequency at ω_c .

Important Note: In figure 8.13, oscillations or ringing takes place near band-edge (ω_c) of the filter. These oscillations or ringing is generated because of sidelobes in the frequency response $W(\omega)$ of the window function. This oscillatory behaviour (i.e., ringing effect) near the band edge of the filter is known as Gibbs phenomenon. Thus, the ringing effect takes place because of sidelobes in $W(\omega)$. These sidelobes are generated because of abrupt discontinuity (in case of rectangular window) of the window function. In case of rectangular window, the sidelobes are larger in size because the discontinuity is abrupt. Therefore, ringing effect is maximum in rectangular window.

Therefore, different window functions are developed which consists of taper and decays gradually towards zero. This reduces side lobes and hence ringing effect in $H(\omega)$.

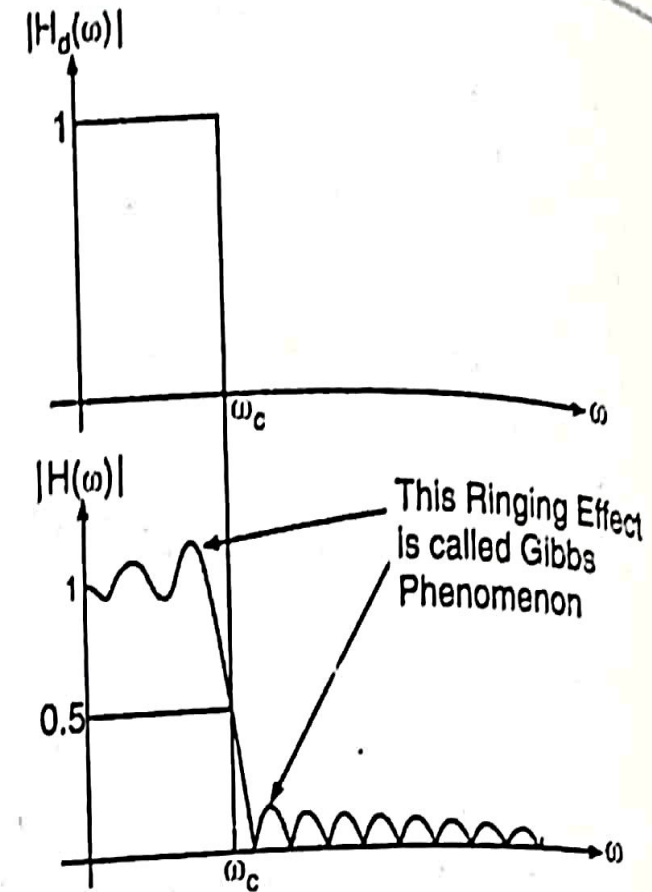


Fig. 8.13. (a) The desired frequency response $H_d(\omega)$
(b) The frequency response of FIR filter obtained by windowing. It has smoothing and ringing effect because of windowing

8.14 FIR FILTER DESIGN USING KAISER WINDOW

By using Kaiser window, it is quite possible to obtain separate control upon length or order (M) of filter and the transition width of the main lobe.

The magnitude specifications of required FIR filter has been shown in figure 8.14.

Here, δ_p = Peak passband deviation
 δ_s = Stopband deviation
 ω_p = Passband edge frequency
 ω_s = Stopband edge frequency
 Δf = Transition width

$$= \frac{\omega_s - \omega_p}{2\pi}$$

In Kaiser window, there are two main parameters. These are the length ($M + 1$) of window and the shape parameter β . The Kaiser window may be defined as under:

$$w(n) = \begin{cases} \frac{I_0 \left\{ \beta \left[1 - \left(\frac{n - \alpha}{\alpha} \right)^2 \right] \right\}^{\frac{1}{2}}}{I_0(\beta)} & \text{for } 0 \leq n \leq M \\ 0 & \text{otherwise} \end{cases}$$

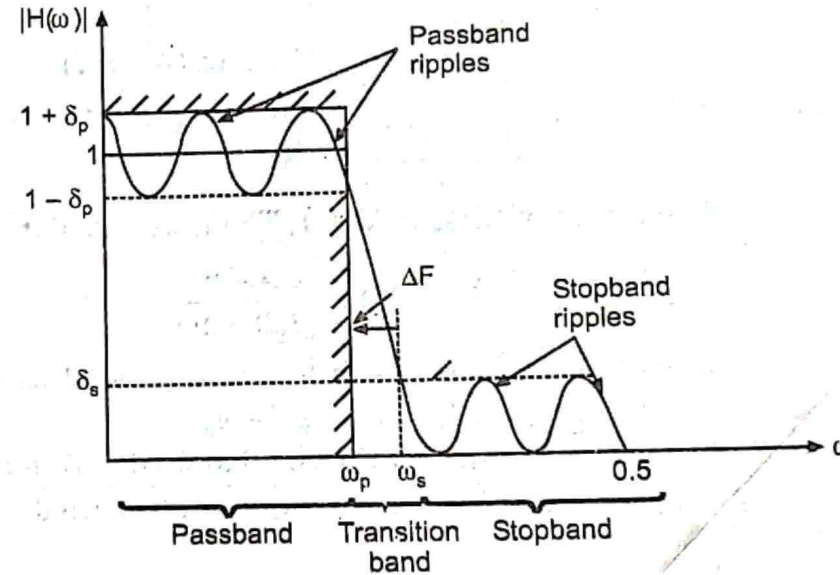


Fig. 8.14. Magnitude specification

Here, $\alpha = \frac{M}{2}$ and $I_0 ()$ is the 0th order modified bessel function of first kind. This function may be obtained by using the following expression:

$$I_0(x) = 1 + \frac{(0.25 x^2)}{(1!)^2} + \frac{(0.25 x^2)^2}{(2!)^2} + \frac{(0.25 x^2)^3}{(3!)^2} + \dots$$

Designing steps

(i) First, we choose the optimum value of ripple ' δ ' using the equation,

$$\delta = \text{Minimum of } \delta_p \text{ and } \delta_s$$

Here,

$$\delta_p = 10^{-0.05} A_p$$

or, we obtain δ_p by solving the following expression :

$$A_p = 20 \log_{10} \left(\frac{1 + \delta_p}{1 - \delta_p} \right)$$

$$\text{and } \delta_s = \frac{10^{0.05} A_p - 1}{10^{0.05} A_p + 1}$$

or, we obtain δ_s by solving the following expression:

$$A_s = -20 \log_{10} (\delta_s)$$

(ii) Then, we calculate attenuation 'A' in dB using the following expression :

$$A = -20 \log_{10} (\delta)$$

(iii) Then, we calculate the parameter β from empirical equations proposed by Kaiser as under:

$$\beta = \begin{cases} 0.1102(A - 8.7) & \text{for } A > 50 \\ 0.5842(A - 21)^{0.4} + 0.07886(A - 21) & \text{for } 21 \leq A \leq 50 \\ 0 & \text{for } A < 21 \end{cases}$$

(iv) The length of filter is $M + 1$. Then, we calculate value of M using the following expression:

$$M = \frac{A - 8}{2.285 \Delta\omega}$$

Here, $\Delta\omega$ = Transition width and is expressed as under :

$$\Delta\omega = \omega_s - \omega_p$$

Where ω_s = Stopband edge frequency and ω_p = Passband edge frequency.

(v) Now, we select the desired impulse response depending on type of filter e.g., low pass, high pass etc.

(vi) Finally, we determine the FIR filter coefficients using the relation $h(n) = h_d(n) \cdot W(n)$.

EXAMPLE 8.9 Design the symmetric FIR low pass filter for which desired frequency response is expressed as

$$H_d(\omega) = \begin{cases} e^{-j\omega\tau} & \text{for } |\omega| \leq \omega_c \\ 0 & \text{elsewhere} \end{cases} \quad \dots(i)$$

The length of the filter should be 7 and $\omega_c = 1$ radians/sample. Make use of rectangular window.

Solution: Given that

Desired frequency response is $H_d(\omega)$

Length of the filter i.e., $M = 7$

Cut-off frequency $\omega_c = 1$ radians/sample

We know that the desired unit sample response is given by:

$$h_d(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} H_d(\omega) e^{j\omega n} d\omega \quad \dots(ii)$$

According to equation (i), given frequency response $H_d(\omega)$ is

$$H_d(\omega) = \begin{cases} e^{-j\omega\tau} & \text{for } -1 \leq \omega \leq 1 \text{ since } \omega_c = 1 \\ 0 & \text{elsewhere} \end{cases}$$

Therefore, equation (ii) will be given by

$$h_d(n) = \frac{1}{2\pi} \int_{-1}^1 e^{-j\omega\tau} \cdot e^{j\omega n} d\omega = \frac{1}{2\pi} \int_{-1}^1 e^{j\omega(n-\tau)} d\omega \quad \dots(iii)$$

or

$$h_d(n) = \frac{1}{2\pi} \left[\frac{e^{j\omega(n-\tau)}}{j(n-\tau)} \right]_{-1}^1 = \frac{\sin(n-\tau)}{\pi(n-\tau)} \text{ for } n \neq \tau$$

When $n = \tau$, equation (iii) will become

$$h_d(n) = \frac{1}{2\pi} \int_{-1}^1 d\omega = \frac{1}{\pi}$$

Therefore, we obtain $h_d(n)$ as under:

$$h_d(n) = \begin{cases} \frac{\sin(n-\tau)}{\pi(n-\tau)} & \text{for } n \neq \tau \\ \frac{1}{\pi} & \text{for } n = \tau \end{cases} \quad \dots(iv)$$

Now, let us determine the value of τ .

We know that the filter is symmetric, therefore, we have

$$h(n) = h(M-1-n)$$

Further, we know that $h(n) = h_d(n) \cdot w(n)$, thus, above condition becomes:

$$h_d(n) w(n) = h_d(M-1-n) w(n)$$

or

$$h_d(n) = h_d(M-1-n)$$

Therefore, using equation (iv), we get

$$\frac{\sin(n-\tau)}{\pi(n-\tau)} = \frac{\sin(M-1-n-\tau)}{\pi(M-1-n-\tau)}$$

The above condition will be satisfied if, we have

$$-(n-\tau) = M-1-n-\tau$$

or
$$\tau = \frac{M-1}{2} \quad \dots(v)$$

Therefore, $h_d(n)$ in equation (iv) becomes,

$$h_d(n) = \begin{cases} \frac{\sin\left(n - \frac{M-1}{2}\right)}{\pi\left(n - \frac{M-1}{2}\right)} & \text{for } n \neq \frac{M-1}{2} \\ \frac{1}{\pi} & \text{for } n = \frac{M-1}{2} \end{cases} \quad \dots(vi)$$

With $M = 7$, the last equation becomes,

$$h_d(n) = \begin{cases} \frac{\sin(n-3)}{\pi(n-3)} & \text{for } n \neq 3 \\ \frac{1}{\pi} & \text{for } n = 3 \end{cases} \quad \dots(vii)$$

Table 8.4. Calculation of $h_d(n)$

S.No.	n	Value of Co-efficient $h_d(n)$ According to Equation (vii)
1.	0	$h_d(0) = \frac{\sin(-3)}{-3\pi} = 0.01497$
2.	1	$h_d(1) = \frac{\sin(-2)}{-2\pi} = 0.14472$
3.	2	$h_d(2) = \frac{\sin(-1)}{-\pi} = 0.26785$
4.	3	$h_d(3) = \frac{1}{\pi} = 0.31831$
5.	4	$h_d(4) = \frac{\sin(1)}{\pi} = 0.26785$
6.	5	$h_d(5) = \frac{\sin(2)}{2\pi} = 0.14472$
7.	6	$h_d(6) = \frac{\sin(3)}{3\pi} = 0.01497$

Now, let us obtain $h(n)$ by windowing operation.

Here, it may be noted that we have calculated only 7 values of $h_d(n)$. However, according to equation (vii), we can calculate infinite number of values of $h_d(n)$. Because we are using a window of '7' values, we will require only seven values of $h_d(n)$, i.e.,

$$h(n) = h_d(n) \cdot w(n)$$

We know that for rectangular window, we have

$$w(n) = \begin{cases} 1 & \text{for } 0 \leq n \leq 6 \\ 0 & \text{elsewhere} \end{cases}$$

Therefore,

$$h(n) = \begin{cases} h_d(n) & \text{for } 0 \leq n \leq 6 \\ 0 & \text{elsewhere} \end{cases}$$

Hence, from Table 8.4, co-efficients of FIR filter will become:

$$\begin{aligned} h(0) &= 0.01497 & h(1) &= 0.14472 \\ h(2) &= 0.26785 & h(3) &= 0.31831 \\ h(4) &= 0.26785 & h(5) &= 0.14472 \\ h(6) &= 0.01497 \end{aligned}$$

This is the unit sample response of required FIR filter. Because, the filter is symmetric, above co-efficients satisfy the condition of $h(n) = h(M-1-n)$ for $M = 7$, i.e.,

$$h(n) = h(6-n)$$

This means that

$$h(0) = h(6)$$

$$h(1) = h(5)$$

$$h(2) = h(4)$$

EXAMPLE 8.10 Design the FIR filter of example 8.9 making use of Hanning window function.

Solution: We know that Hanning window function is given by following expression:

$$w(n) = \frac{1}{2} \left(1 - \cos \frac{2\pi n}{M-1} \right) \quad \dots(i)$$

Table 8.5 presents the values of window function calculated according to equation (i)

Table 8.5. Calculation of Co-efficients for Hanning Window for $M = 7$

S.No	n	$w(n) = \frac{1}{2} \left(1 - \cos \frac{2\pi n}{M-1} \right), M = 7$
1.	0	$w(0) = \frac{1}{2} [1 - \cos(0)] = 0$
2.	1	$w(1) = \frac{1}{2} \left[1 - \cos \left(\frac{2\pi}{6} \right) \right] = \frac{1}{4}$
3.	2	$w(2) = \frac{1}{2} \left[1 - \cos \left(\frac{4\pi}{6} \right) \right] = \frac{3}{4}$
4.	3	$w(3) = \frac{1}{2} \left[1 - \cos \left(\frac{6\pi}{6} \right) \right] = 1$
5.	4	$w(4) = \frac{1}{2} \left[1 - \cos \left(\frac{8\pi}{6} \right) \right] = \frac{3}{4}$
6.	5	$w(5) = \frac{1}{2} \left[1 - \cos \left(\frac{10\pi}{6} \right) \right] = \frac{1}{4}$
7.	6	$w(6) = \frac{1}{2} \left[1 - \cos \left(\frac{12\pi}{6} \right) \right] = 0$

Now, let us use the values of $h_d(n)$ calculated in previous example. We know that in window method, $h(n)$ is given by:

$$h(n) = h_d(n) w(n)$$

Table 8.6 shows the calculation of $h(n)$. Here, $h_d(n)$ of table 8.4 is used and $w(n)$ of Table 8.5 is used.

Table 8.6. Calculation of $h(n)$

S.No	n	$h_d(n)$	$w(n)$	$h(n) = h_d(n) w(n)$
1.	0	0.01497	0	$h(0) = 0$
2.	1	0.14472	$\frac{1}{4}$	$h(1) = 0.03618$
3	2	0.26785	$\frac{3}{4}$	$h(2) = 0.20089$
4.	3	0.31831	1	$h(3) = 0.31831$
5.	4	0.26785	$\frac{3}{4}$	$h(4) = 0.20089$
6.	5	0.14472	$\frac{1}{4}$	$h(5) = 0.03618$
7.	6	0.01497	0	$h(6) = 0$

In above table, it may also be observed that unit sample response is symmetric and it satisfies the condition of $h(n) = h(M-1-n)$.

Comments

In example 8.9, we have calculated value of τ as under:

$$\tau = \frac{M-1}{2}$$

The $H_d(\omega)$ of equation (i) in Example 8.9 can be given by:

$$H_d(\omega) = \begin{cases} e^{-j\omega\left(\frac{M-1}{2}\right)} & \text{for } |\omega| \leq \omega_c \\ 0 & \text{elsewhere} \end{cases} \quad \dots(ii)$$

The time shifting property of Fourier transform states that,

$$\text{FT}\{x(n-k)\} = e^{-j\omega k} X(\omega)$$

This means that multiplication by $e^{-j\omega k}$ to the Fourier transform is equivalent to delaying the time domain sequence by 'k' units. Therefore, in equation (ii), it may be observed that $H_d(\omega)$ has magnitude of 1 for $|\omega| < \omega_c$.

And, multiplication by $e^{-j\omega\left(\frac{M-1}{2}\right)}$ implies delaying $h_d(n)$ by $\frac{M-1}{2}$ samples.

Hence, we can write magnitude response $|H_d(\omega)|$ and phase response $\angle H_d(\omega)$ from equation (ii) as under:

$$|H_d(\omega)| = \begin{cases} 1 & \text{for } |\omega| \leq \omega_c \\ 0 & \text{elsewhere} \end{cases}$$

and
$$\angle H_d(\omega) = -\omega\left(\frac{M-1}{2}\right)$$

Figure 8.16 illustrates the magnitude and phase response of the desired filter.

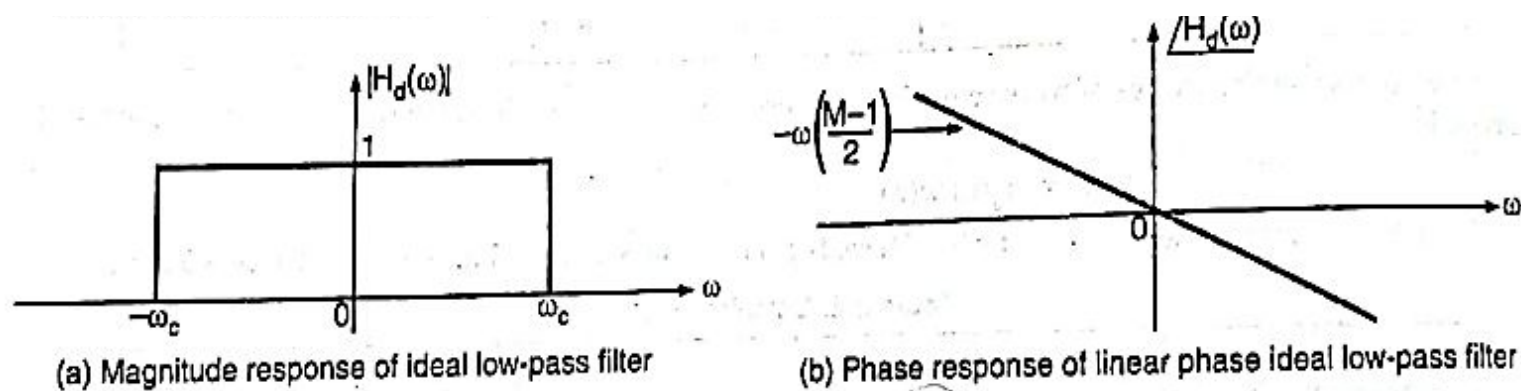


Fig. 8.16.

Figure 8.16 illustrates that $H_d(\omega)$ represents ideal low-pass filter. Hence, in Example 8.9 we have designed ideal low-pass filter using rectangular window. But, because of windowing (i.e., truncation of $h_d(n)$), the frequency response of FIR filter, i.e., $H(\omega)$ is no more ideal.

Important Point: Here, we started from desired frequency response $H_d(\omega)$. Then, we obtained $h_d(n)$ by inverse fourier transform of $H_d(\omega)$. Then, $h(n)$ is obtained by windowing of $h_d(n)$. This is also known as FIR filter design using inverse Fourier transform, because $h_d(n)$ is obtained by inverse Fourier transform. However, windowing is more commonly used name for this method.

EXAMPLE 8.11 Design a low-pass digital filter to be used in an A/D - H(z) - D/A structure that will have - 3 dB cut-off at 30π rad/sec and an attenuation of 50 dB at 45π rad/sec. The filter is required to have a linear phase and the system uses a sampling rate of 100 samples/second.

Solution: Given that

3 dB cut-off frequency at 30π rad/sec, i.e.,

$$\Omega_c = 30 \pi \text{ rad/sec}$$

Sampling frequency $F_{SF} = 100 \text{ Hz.}$

Stopband attenuation of 50 dB at 45π rad / sec, i.e.,

$$A_s = 50 \text{ dB for } \Omega_s = 45 \pi \text{ rad / sec}$$

First, let us obtain specifications of equivalent digital filter.

We know that $\omega = 2\pi f$, and $f = \frac{F}{F_{SF}}$, therefore, we write,

$$\omega = 2\pi \frac{F}{F_{SF}}$$

Because $\Omega = 2\pi F$, above expression becomes

$$\omega = \frac{\Omega}{F_{SF}}$$

Let

$$\Omega_c = \Omega_1 = 30\pi \text{ rad / sec, and } \omega$$

$$\Omega_s = \Omega_2 = 45\pi \text{ rad / sec}$$

Therefore, we have

$$\omega_1 = \frac{\Omega_1}{F_{SF}} = \frac{30\pi}{100} = 0.3\pi \text{ rad/sample } \omega_c$$

and

$$\omega_2 = \frac{\Omega_2}{F_{SF}} = \frac{45\pi}{100} = 0.45\pi \text{ rad /sample } \omega_s$$

Hence, we have the specifications of equivalent digital filter as under:

$$3 \text{ dB attenuation at } \omega_1 = 0.3 \pi \text{ rad / sample}$$

$$A_s = 50 \text{ dB at } \omega_2 = 0.45 \pi \text{ rad/sample}$$

...(i)

Now, let us select the type of window.

Here, note that minimum stopband attenuation given is 50 dB. From Table 8.2, it may be observed that hanning window provides - 44 dB of stopband attenuation and hamming window provides - 53 dB of stopband attenuation. Therefore, we should use hamming window to obtain the desired attenuation of 50 dB. Therefore, window to be used: **Hamming window**.

Now, let us determine the order of the filter

We have selected hamming window for the given filter. From Table 8.2, the width of the main lobe of this window is $\frac{8\pi}{M}$. Thus, we have

$$k \left(\frac{2\pi}{M} \right) = \frac{8\pi}{M} \approx$$

or

$$k = 4$$

The order of the filter is given as under:

$$N = k \left(\frac{2\pi}{\omega_2 - \omega_1} \right) \quad \frac{2\pi}{\omega_s - \omega_c}$$

Substituting values in above equation, we get

$$N = 4 \left(\frac{2\pi}{0.45\pi - 0.3\pi} \right) = 53.33$$

Here, let us design the linear phase FIR filter of odd length. Hence, let us select next odd integer after 53.33, i.e., $N = 55$.

Therefore, length of the filter, $M = 55$

Now, let us obtain desired unit sample response $h_d(n)$

Let us consider that the filter has ideal response as shown in figure 8.16. It is given by

$$H_d(\omega) = \begin{cases} e^{-j\omega \left(\frac{M-1}{2} \right)} & \text{for } -\omega_c \leq \omega \leq \omega_c \\ 0 & \text{otherwise} \end{cases} \quad \dots(ii)$$

This is ideal low-pass filter having cut-off frequency ω_c and linear phase response of $-\omega\left(\frac{M-1}{2}\right)$ as shown in figure 8.16.

The desired unit sample response $h_d(n)$ may be obtained by taking inverse Fourier transform of $H_d(\omega)$.

$$\text{This means that } h_d(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} H_d(\omega) e^{j\omega n} d\omega$$

Substituting for $H_d(\omega)$ from equation (ii) in above equation, we get

$$h_d(n) = \frac{1}{2\pi} \int_{-\omega_c}^{\omega_c} e^{-j\omega\left(\frac{M-1}{2}\right)} e^{j\omega n} d\omega = \frac{1}{2\pi} \int_{-\omega_c}^{\omega_c} e^{j\omega\left(n - \frac{M-1}{2}\right)} d\omega \quad \dots(iii)$$

or

$$h_d(n) = \frac{1}{2\pi} \left[\frac{e^{j\omega\left(n - \frac{M-1}{2}\right)}}{j\left(n - \frac{M-1}{2}\right)} \right]_{-\omega_c}^{\omega_c} = \frac{1}{2\pi} \left[\frac{e^{j\omega_c\left(n - \frac{M-1}{2}\right)} - e^{-j\omega_c\left(n - \frac{M-1}{2}\right)}}{j\left(n - \frac{M-1}{2}\right)} \right]$$

or

$$h_d(n) = \frac{\sin\left[\omega_c\left(n - \frac{M-1}{2}\right)\right]}{\pi\left(n - \frac{M-1}{2}\right)} \quad \text{for } n \neq \frac{M-1}{2}$$

when $n = \frac{M-1}{2}$, equation (iii) becomes,

$$h_d(n) = \frac{1}{2\pi} \int_{-\omega_c}^{\omega_c} d\omega = \frac{1}{2\pi} \left[\omega \right]_{-\omega_c}^{\omega_c} = \frac{\omega_c}{\pi}$$

Therefore, we obtain $h_d(n)$ as under:

$$h_d(n) = \begin{cases} \frac{\sin\left[\omega_c\left(n - \frac{M-1}{2}\right)\right]}{\pi\left(n - \frac{M-1}{2}\right)} & \text{for } n \neq \frac{M-1}{2} \\ \frac{\omega_c}{\pi} & \text{for } n = \frac{M-1}{2} \end{cases} \quad \dots(iv)$$

Now, let us obtain $h(n)$ by windowing:

In this example, we have selected hamming window to get the required attenuation. From Table 8.3, the equation of hamming window is expressed as:

$$w(n) = 0.54 - 0.46 \cos\left(\frac{2\pi n}{M-1}\right) \text{ for } 0 \leq n \leq M-1$$

$h(n)$ is given by $h(n) = h_d(n) w(n)$

Substituting the respective expressions in above equation, we obtain

$$h(n) = \begin{cases} \frac{\sin\left[\omega_c\left(n - \frac{M-1}{2}\right)\right]}{\pi\left(n - \frac{M-1}{2}\right)} \cdot \left[0.54 - 0.46 \cos\left(\frac{2\pi n}{M-1}\right)\right] & \text{for } n \neq \frac{M-1}{2} \\ \frac{\omega_c}{\pi} \left[0.54 - 0.46 \cos\left(\frac{2\pi n}{M-1}\right)\right] & \text{for } n = \frac{M-1}{2} \end{cases}$$

We have, $M = 55$, hence $\frac{M-1}{2} = 27$. Similarly, from equation (i), $\omega_1 = \omega_c = 0.3 \pi \text{ rad/sample}$.

Substituting these values in above equation, we obtain

$$h(n) = \begin{cases} \frac{\sin[0.3\pi(n-27)]}{\pi(n-27)} \left[0.54 - 0.46 \cos\left(\frac{\pi n}{27}\right) \right] & \text{for } n \neq 27 \\ 0.3 \left[0.54 - 0.46 \cos\left(\frac{\pi n}{27}\right) \right] & \text{for } n = 27 \end{cases}$$

for $n = 0$ to 54

Important Point: The values of unit sample response may be obtained above equation to get desired linear phase FIR filter.

EXAMPLE 8.12 Design a normalised linear phase FIR filter having the phase delay of $\tau = 4$ and at least 40 dB attenuation in the stopband. Also, obtain the magnitude/frequency response of the filter.

Solution: First, let us obtain ω_c

The linear phase FIR filter is normalised. This means cut off frequency will be:

$$\omega_c = 1 \text{ rad/sample}$$

Now, let us obtain length M of the filter

Phase delay, $\tau = 4$. We know that for a linear phase FIR filter, we have

$$\tau = \frac{M-1}{2}$$

Now, $\tau = 4$ gives, $4 = \frac{M-1}{2}$ or $M = 9$

Therefore, the length of the filter is $M = 9$.

Now, let us choose the type of the window.

At least, 40 dB attenuation is required in stopband. From Table 8.2, it is obvious that Hanning window provides minimum of 44 dB attenuation.

Therefore,

Window used: *Hanning window*

Now, let us obtain desired unit sample response $h_d(n)$.

In example 8.11, we have observed that $h_d(n)$ has been obtained for an ideal low-pass filter of cut-off frequency ω_c . Such $h_d(n)$ is given by equation (iv) in previous example as under :

$$h_d(n) = \begin{cases} \frac{\sin \left[\omega_c \left(n - \frac{M-1}{2} \right) \right]}{\pi \left(n - \frac{M-1}{2} \right)} & \text{for } n \neq \frac{M-1}{2} \\ \frac{\omega_c}{\pi} & \text{for } n = \frac{M-1}{2} \end{cases}$$

In above expression, we have to substitute $\omega_c = 1$ rad/sample and $\frac{M-1}{2} = \tau = 4$. This means that

$$h_d(n) = \begin{cases} \frac{\sin(n-4)}{\pi(n-4)} & \text{for } n \neq 4 \\ \frac{1}{\pi} & \text{for } n = 4 \end{cases} \quad \dots(i)$$

Now, let us obtain $h(n)$ by windowing.

In this example, we have selected Hanning window.

From Table 8.3, Hanning window is given as under:

$$w(n) = \frac{1}{2} \left(1 - \cos \frac{2\pi n}{M-1} \right)$$

Substituting for M, above expression will take the following form:

$$w(n) = \frac{1}{2} \left(1 - \cos \frac{\pi n}{4} \right)$$

Also, $h(n)$ may be obtained by windowing as under:

$$h(n) = h_d(n) \cdot w(n)$$

$$\text{or } h(n) = \begin{cases} \frac{\sin(n-4) \cdot \left(1 - \cos \frac{\pi n}{4}\right)}{2\pi(n-4)} & \text{for } n \neq 4 \\ \frac{1 - \cos \frac{\pi n}{4}}{2\pi} & \text{for } n = 4 \end{cases} \quad \dots(ii)$$

for $n = 0$ to 8

This is the unit sample response of the desired linear phase FIR filter.

Now, let us obtain the magnitude (frequency) response.

Here, order of the filter is, $M = 9$ i.e., odd. For odd M, the magnitude/frequency response of FIR filters is given by:

$$|H(\omega)| = h\left(\frac{M-1}{2}\right) + 2 \sum_{n=0}^{\frac{M-3}{2}} h(n) \cos \omega \left(n - \frac{M-1}{2} \right)$$

Substituting for $M = 9$, above expression becomes,

$$|H(\omega)| = h(4) + 2 \sum_{n=0}^3 h(n) \cos \omega(n-4)$$

Values of $h(n)$ can be substituted in above expression from equation (ii). Then $|H(\omega)|$ may be obtained for various values of ω .

Now, let us obtain magnitude response
 Here, length of the unit sample response is $M = 7$, i.e., odd.
 For odd M , the magnitude response is given by:

$$|H(\omega)| = h\left(\frac{M-1}{2}\right) + 2 \sum_{n=0}^{\frac{M-1}{2}-1} h(n) \cos \omega \left(n - \frac{M-1}{2}\right)$$

$$\begin{aligned} \text{For } M = 7, |H(\omega)| &= h(3) + 2 \sum_{n=0}^2 h(n) \cos \omega (n-3) \\ &= h(3) + 2 [h(0) \cos \omega (-3) + h(1) \cos \omega (1-3) + h(2) \cos \omega (2-3)] \\ &= h(3) + 2 [h(0) \cos (3\omega) + h(1) \cos (2\omega) + h(2) \cos (\omega)] \end{aligned}$$

Substituting values in above expression, we shall have

$$\begin{aligned} |H(\omega)| &= 0.31831 + 2[-0.04462 \cos (3\omega) - 0.26517 \cos (2\omega) + 0.02159 \cos (\omega)] \\ &= 0.31831 - 0.08924 \cos (3\omega) - 0.53034 \cos (2\omega) + 0.04318 \cos (\omega) \end{aligned}$$

The frequency response may be obtained by substituting $-\pi \leq \omega \leq \pi$ in the above expression.

EXAMPLE 8.14 Design an FIR linear phase filter using Kaiser window to meet the following specifications:

$$\begin{aligned} 0.99 &\leq |H(e^{j\omega})| \leq 1.01, \text{ for } 0 \leq |\omega| \leq 0.19\pi \\ |H(e^{j\omega})| &\leq 0.01, \text{ for } 0.21\pi \leq |\omega| \leq \pi \end{aligned}$$

Solution: The given specifications may be rewritten as under:

$$\begin{aligned} 1 - 0.01 &\leq |H(e^{j\omega})| \leq 1 + 0.01, \text{ for } 0 \leq |\omega| \leq 0.19\pi \\ |H(e^{j\omega})| &\leq 0.01, \text{ for } 0.21\pi \leq |\omega| \leq \pi \end{aligned}$$

On comparing the above given specifications with equation (8.32), we find that

$$\begin{aligned} \delta_1 &= 0.01, \quad \delta_2 = 0.01 \\ \omega_p &= 0.19\pi, \quad \omega_s = 0.21\pi \end{aligned}$$

Therefore, transition width $\Delta\omega$ would be:

$$\begin{aligned} \Delta\omega &= \omega_s - \omega_p \\ \text{or } \Delta\omega &= 0.21\pi - 0.19\pi = 0.02\pi \end{aligned}$$

And the minimum value of ripple is given by:

$$\delta = \text{minimum of } \delta_1 \text{ and } \delta_2$$

Therefore, $\delta = 0.01$

Using equation (8.51), the value of A may be calculated as under:

$$A = -20 \log_{10} \delta = -20 \log_{10} 0.01 = 40$$

Now, let us determine cut-off frequency ω_c

The cut-off frequency of $H_d(\omega)$ is required. We are given passband edge frequency ω_p and stopband edge frequency ω_s .

For such characteristic, cut-off frequency is given by:

$$\omega_c = \frac{\omega_p + \omega_s}{2} \quad \dots (ii)$$

Substituting the values in above expression, we get

$$\omega_c = \frac{0.19\pi + 0.21\pi}{2} = 0.2\pi$$

Now, let us obtain values of β and M.

Here, $A = 40$, which lies in the range of 21 to 50. Hence, β may be obtained as under:

$$\beta = 0.5842 (A - 21)^{0.4} + 0.07886 (A - 21)$$

or
$$\beta = 0.5842 (40 - 21)^{0.4} + 0.07886 (40 - 21)$$

or
$$\beta = 3.395$$

We know that 'M' is given by

$$M = \frac{A - 8}{2.285 \Delta \omega} = \frac{40 - 8}{2.285(0.02\pi)} = 222.88$$

Here, we select the value of M as 223. It may be noted that M is very large since transition band is very small, i.e., 0.02π .

Now, let us obtain the expression of Kaiser window.

We know that $\alpha = \frac{M}{2} = \frac{223}{2} = 111.5$

and $\beta = 3.395$

Substituting the values in equation (8.47), the window will be defined as under:

$$w(n) = \begin{cases} \frac{I_0 \left\{ 3.395 \left[1 - \left(\frac{n - 111.5}{111.5} \right)^2 \right]^{\frac{1}{2}} \right\}}{I_0(3.395)} & \text{for } 0 \leq n \leq 223 \\ 0 & \text{elsewhere} \end{cases}$$

Handwritten note: $w(n) = \frac{I_0(\beta)}{I_0(\beta)}$

The function $I_0()$ in the above expression may be calculated with the help of equation (8.48). Since, it is an infinite series, maximum possible terms must be taken to reduce the error.

Now, let us obtain $h_d(n)$.

This is low-pass type of filter. The ideal desired frequency response is given by:

$$H_d(\omega) = \begin{cases} e^{-j\omega \left(\frac{M-1}{2} \right)} & \text{for } -\omega_c \leq \omega \leq \omega_c \\ 0 & \text{otherwise} \end{cases}$$

Now, $h_d(n)$ may be obtained by taking inverse Fourier transform of above equation, i.e.,

$$h_d(n) = \begin{cases} \frac{\sin \left[\omega_c \left(n - \frac{M-1}{2} \right) \right]}{\pi \left(n - \frac{M-1}{2} \right)} & \text{for } n \neq \frac{M-1}{2} \\ \frac{\omega_c}{\pi} & \text{for } n = \frac{M-1}{2} \end{cases}$$

At the beginning, it was stated that length of the filter is $M + 1$. The above equation is derived for filter length M . Hence, we should replace M by ' $M + 1$ ' in above equation.

This means that

$$h_d(n) = \begin{cases} \frac{\sin \left[\omega_c \left(n - \frac{M}{2} \right) \right]}{\pi \left(n - \frac{M}{2} \right)} & \text{for } n \neq \frac{M}{2} \\ \frac{\omega_c}{\pi} & \text{for } n = \frac{M}{2} \end{cases}$$

Here, $M = 223$, hence, $\frac{M}{2} = 111.5$. Therefore, $h_d(n)$ cannot be calculated for $n = \frac{M}{2}$ since 111.5 is not an integer.

In other words, filter length is $M + 1 = 223 + 1 = 224$, i.e., even.

Therefore, $h_d(n)$ will be given by:

$$h_d(n) = \frac{\sin \left[\omega_c \left(n - \frac{M}{2} \right) \right]}{\pi \left(n - \frac{M}{2} \right)} \quad \text{for } 0 \leq n \leq m$$

Substituting for $\omega_c = 0.2 \pi$ and $M = 223$, the last expression becomes,

$$h_d(n) = \frac{\sin \left[0.2\pi \left(n - \frac{223}{2} \right) \right]}{\pi \left(n - \frac{223}{2} \right)}$$

or
$$h_d(n) = \frac{\sin [0.2\pi (n - 111.5)]}{\pi (n - 111.5)} \quad \text{for } 0 \leq n \leq 223$$

Here, it may be noted that above expression also stands for linear phase requirement. This expression gives desired unit sample response of low-pass filter.

Lastly, let us obtain $h(n)$.

The unit sample response of FIR filter may be obtained by windowing, i.e.,

$$h(n) = h_d(n) w(n)$$

or
$$h(n) = \begin{cases} \frac{\sin [0.2\pi (n - 111.5)]}{\pi (n - 111.5)} \cdot \frac{I_0 \left\{ 3.395 \left[1 - \left(\frac{n - 111.5}{111.5} \right)^2 \right]^{\frac{1}{2}} \right\}}{I_0(3.395)} & \text{for } 0 \leq n \leq 223 \\ 0 & \text{otherwise} \end{cases}$$

This is the required relationship for unit sample response of FIR filter using Kaiser window.